

RecruitAI

AI-Powered Recruitment & Hiring Platform

n8n Capstone Project | Summer School'26

Workflow Documentation

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Domain: Human Resources — Recruitment & Hiring Automation

Core Stack: n8n • Google Forms • Google Sheets • Google Drive • Google Gemini • Gmail

Project scope

This document describes the implemented RecruitAI automation platform, its four independent n8n workflows, workflow interaction, data flow, AI processing, human approval step, integrations, and implementation summary.

Implementation: 4 independent workflows • 20 n8n nodes

1. System Architecture

RecruitAI is implemented as a modular recruitment automation platform. Candidate information is collected through the application process and maintained in a centralized Google Sheets database. Four independent n8n workflows process the candidate lifecycle through registration, AI resume analysis, HR decision, and interview communication.

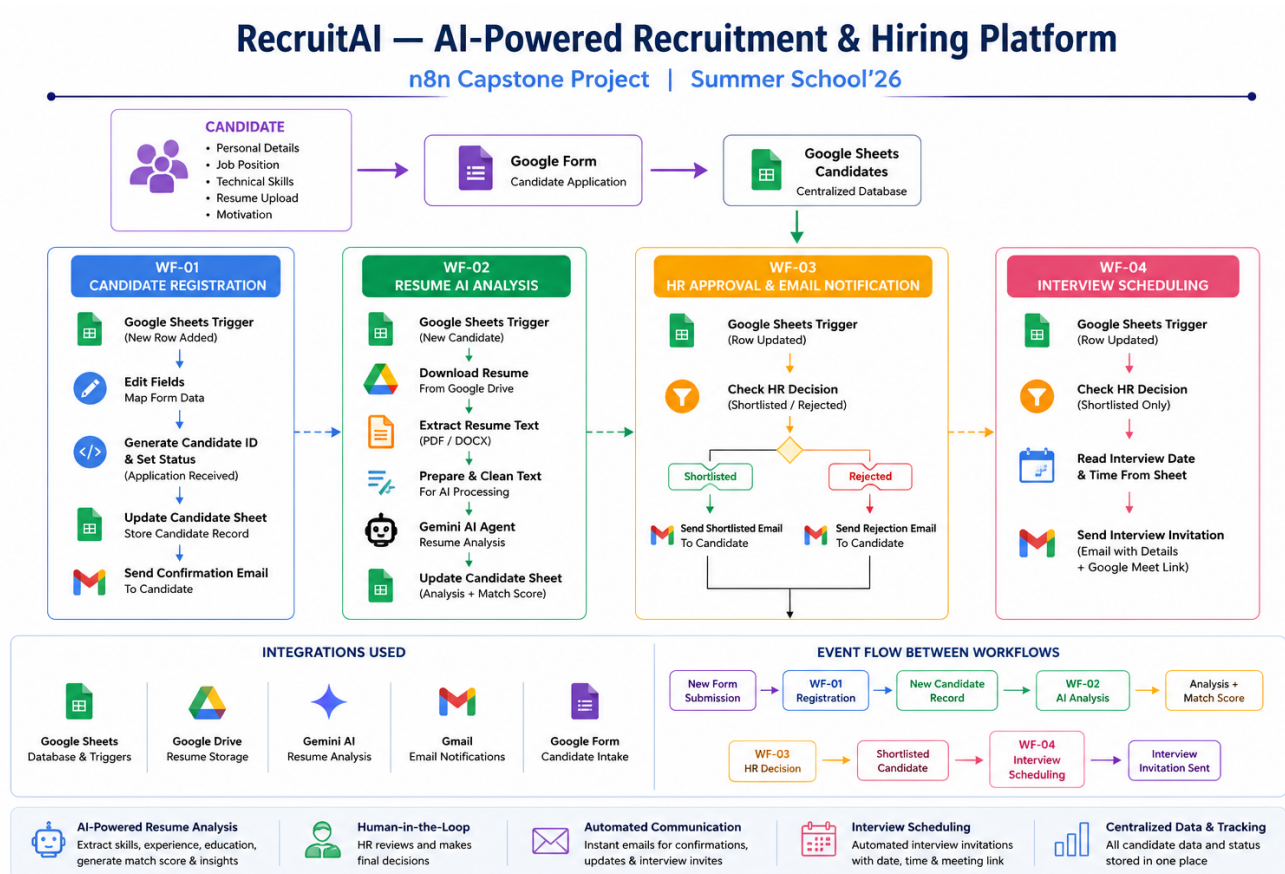


Figure 1. Overall RecruitAI architecture and event flow.

Workflow interaction

The workflows interact through updates to the centralized candidate record. WF-01 creates/updates the initial candidate record, WF-02 stores AI analysis and match score, WF-03 uses the HR Decision field to communicate the outcome, and WF-04 uses the shortlisted state plus interview date/time to send the interview invitation.

Overall event flow: Candidate Application → WF-01 Registration → Candidate Record → WF-02 AI Analysis → HR Decision → WF-03 Notification → WF-04 Interview Scheduling

2. Problem Analysis

Business context

Growing technology companies may receive many applications across different sources. Manual resume screening, candidate record maintenance, status communication, and interview coordination create repetitive administrative work and can delay the recruitment process. RecruitAI addresses these tasks with an automated, modular workflow while keeping the final recruitment decision with HR.

Stakeholders

Stakeholder	Role
Candidates	Submit applications and receive automated recruitment communication.
HR Team	Reviews AI-generated candidate information and makes the final recruitment decision.
Recruiters	Manage candidate records, recruitment status and interview communication.
Hiring Managers	Use structured candidate information during recruitment.
Company Management	Benefits from a more structured and automated recruitment process.

Pain points

- Manual candidate data processing and record maintenance.
- Time-consuming resume review and initial evaluation.
- Repetitive status communication with candidates.
- Manual coordination of interview information.
- Difficulty maintaining a consistent, centralized recruitment record.

Objectives

- Automate candidate registration and resume collection.
- Evaluate resumes using AI.
- Reduce manual interview-related effort.
- Improve candidate communication.
- Maintain a centralized recruitment database.

3. Implementation Summary

The implementation satisfies the mandatory architecture requirement of four independent n8n workflows and a minimum of 20 nodes. The exported workflow files contain the actual node configurations used in the project.

Workflow	Purpose	Nodes
WF-01 Candidate Registration	Candidate intake, candidate ID/status processing, database update and confirmation email.	5
My workflow		8
Workflow 3 – HR Approval & Email Notification	HR decision branching and shortlisted/rejection email communication.	4
Workflow 4 - Interview Scheduling	Shortlisted check and interview invitation email.	3
Total	Implemented recruitment automation platform	20

Implementation characteristics

- AI-powered resume analysis using Google Gemini.
- Human-in-the-loop HR decision through the HR Decision field.
- Conditional branching for shortlisted versus non-shortlisted candidates.
- Automated Gmail notifications.
- Centralized candidate data in Google Sheets.

4. Workflow 1 — Candidate Registration & Resume Collection

Purpose

Automates the initial candidate registration process. A newly added candidate record is processed, a candidate ID and initial application status are generated, the candidate database is updated, and a confirmation email is sent.

Trigger

Google Sheets Trigger — Row Added

Input

Candidate application fields including name, email, phone, job position, experience, qualification, skills, motivation and resume information.

Output

Centralized candidate record with Candidate ID and Application Status, plus a candidate confirmation email.

Workflow logic

New row → map candidate fields → generate candidate ID/status → update candidate record → send confirmation.

Implemented nodes

Node	n8n type
Google Sheets Trigger	googleSheetsTrigger
Edit Fields	set
Code in JavaScript	code
Update row in sheet	googleSheets
Send a message	gmail

Node flow

Google Sheets Trigger	Edit Fields	Code in JavaScript	Update row in sheet
Send a message			

5. Workflow 2 — Resume Parsing & AI Evaluation

Purpose

Retrieves the candidate resume, extracts its content, prepares the AI input, uses Google Gemini to analyze the resume, and stores the resulting analysis and AI match score in the candidate database.

Trigger

Google Sheets Trigger — Candidate record processing

Input

Candidate record and resume link/file.

Output

Resume Analysis and AI Match Score stored in the candidate record, along with selected structured candidate information.

Workflow logic

Candidate record → download resume → extract text → prepare AI input → Gemini AI Agent → update candidate record.

Implemented nodes

Node	n8n type
Google Sheets Trigger	googleSheetsTrigger
Download file	googleDrive
Edit Fields	set
Extract from File	extractFromFile
Edit Fields1	set
AI Agent	@n8n/n8n-nodes-langchain.agent
Google Gemini Chat Model	@n8n/n8n-nodes-langchain.lmChatGoogleGemini
Update row in sheet	googleSheets

Node flow

Google Sheets Trigger	Download file	Edit Fields	Extract from File
Edit Fields1	AI Agent	Google Gemini Chat Model	Update row in sheet

6. Workflow 3 — HR Approval & Email Notification

Purpose

Handles the human HR decision. The workflow checks the HR Decision field and routes the candidate to a shortlisted or non-shortlisted email branch.

Trigger

Google Sheets Trigger — Row Updated

Input

Candidate record containing HR Decision, Full Name and Email Address.

Output

Automated shortlisted or non-shortlisted email.

Workflow logic

Updated row → check HR Decision → Shortlisted branch OR non-shortlisted branch → Gmail.

Implemented nodes

Node	n8n type
Google Sheets Trigger	googleSheetsTrigger
If	if
Send a message	gmail
Send a message1	gmail

Node flow

Google Sheets Trigger	If	Send a message	Send a message1
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7. Workflow 4 — Interview Scheduling

Purpose

Handles interview communication for shortlisted candidates. It checks the HR Decision field and, when the candidate is shortlisted, sends the interview date, time and meeting information by email.

Trigger

Google Sheets Trigger — Row Updated

Input

Candidate record containing HR Decision, candidate email, interview date and interview time.

Output

Interview invitation email for shortlisted candidates.

Workflow logic

Updated row → check Shortlisted status → read interview details → Gmail interview invitation.

Implemented nodes

Node	n8n type
Google Sheets Trigger	googleSheetsTrigger
If	if
Send a message	gmail

Node flow

Google Sheets Trigger	If	Send a message
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8. Workflow Interaction & Data Flow

The platform uses the centralized candidate database as the shared state between the independent workflows. This allows each workflow to perform a focused task while the candidate record carries the current recruitment information forward.

Stage	Data / State	Automation
1. Application	Candidate details + resume link	WF-01 registration
2. Registration	Candidate ID + Application Status	Candidate record update + confirmation email
3. AI Evaluation	Extracted resume content + structured AI output	WF-02 Gemini analysis + match score
4. HR Review	HR Decision	Human decision recorded in candidate database
5. Decision Communication	Shortlisted / non-shortlisted state	WF-03 Gmail notification
6. Interview	Interview Date + Interview Time	WF-04 interview invitation

End-to-end event flow

Candidate submits application

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WF-01 creates/updates the candidate record

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WF-02 retrieves and analyzes the resume with Gemini

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Resume Analysis + AI Match Score are stored

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HR reviews the candidate and enters HR Decision

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WF-03 sends the appropriate decision email

↓

If shortlisted, WF-04 sends the interview invitation

9. AI Processing & Human-in-the-Loop Design

AI role

Google Gemini is used to process the extracted resume content and generate structured candidate information, resume analysis and an AI match score. The result is written back to the centralized candidate database for HR review.

Human approval

The AI output does not directly make the final hiring decision. HR uses the candidate information and AI-generated evaluation to enter the HR Decision. Workflow 3 then automates the appropriate communication based on that human decision.

Conditional decision logic

Workflow 3 checks whether HR Decision equals "Shortlisted". The shortlisted branch sends a positive status email; the alternate branch sends a non-shortlisted email. Workflow 4 separately checks for the shortlisted state before sending interview information.

AI-to-human flow

Resume → AI analysis → Match Score → HR Review → HR Decision → Automated Communication

10. Technology Stack & Integrations

Technology	Role in RecruitAI
n8n	Workflow orchestration, triggers, data transformation, conditional logic and integrations.
Google Forms	Candidate application intake.
Google Sheets	Centralized candidate database and workflow trigger/state mechanism.
Google Drive	Resume file storage and retrieval.
Google Gemini	AI-powered resume analysis and match scoring.
Gmail	Candidate confirmation, recruitment decision and interview communication.

11. Project Outcomes

- Automates candidate registration and initial record management.
- Automates resume retrieval and AI-powered evaluation.
- Creates a structured AI match score and resume analysis for HR review.
- Keeps HR as the final decision-maker.
- Automates candidate status communication.
- Automates interview invitation communication for shortlisted candidates.
- Maintains a centralized recruitment record in Google Sheets.

12. Advanced Features & Scope

Implemented advanced features

Feature	Status	Implementation
AI-powered decision support	Implemented	Google Gemini analyzes resumes and generates a match score and structured analysis.
Human approval	Implemented	HR Decision is entered by HR and drives the communication workflow.
Conditional branching	Implemented	WF-03 and WF-04 use conditional checks for shortlisted candidates.
Automated notifications	Implemented	Gmail sends confirmation, decision and interview messages.
Error handling / retry logic	Not implemented as a dedicated feature	Can be added in a future version.
Audit/logging workflow	Not implemented as a dedicated feature	Candidate status and timestamps remain in the centralized sheet; a dedicated audit trail can be added later.
Cron/scheduled workflow	Not a separate scheduled workflow	Current workflows use Google Sheets triggers rather than a dedicated Cron workflow.
Webhook workflow	Not implemented	Not required for the current four-workflow implementation.

Future enhancements

- Google Calendar integration for automated interview slot management.
- Automated interview feedback collection.
- Offer letter generation.
- Recruitment analytics and management reporting.
- Duplicate candidate detection.
- Dedicated error handling, retries and audit logging.

13. Final Workflow Summary

Workflow	Primary Function	AI	Human Decision
WF-01 Candidate Registration	Candidate intake, ID/status processing and confirmation	No	No
WF-02 Resume AI Analysis	Resume retrieval, extraction, AI analysis and match score	Yes — Gemini	No
WF-03 HR Approval & Email Notification	HR decision branching and status communication	No	Yes — HR
WF-04 Interview Scheduling	Shortlisted check and interview invitation	No	No

Implementation checklist

- 4 independent n8n workflows — implemented.
- 20 n8n nodes in total — meets the 20–25 node minimum.
- AI-powered resume evaluation — implemented.
- Human HR approval — implemented.
- Conditional branching — implemented.
- Automated Gmail communication — implemented.
- Centralized candidate database — implemented.

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